

Introduction to AI

What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of Computer Science that focuses on creating machines capable of performing tasks that normally require human intelligence.

These tasks include:

- Learning
- Reasoning
- Problem-solving
- Decision making
- Understanding language
- Recognizing images and speech

Definition 1 (John McCarthy, 1956)

"Artificial Intelligence is the science and engineering of making intelligent machines."

John McCarthy is known as the "**Father of Artificial Intelligence**" because he coined the term Artificial Intelligence in 1956.

Definition 2

AI is the study of intelligent agents that perceive their environment and take actions to maximize their chances of achieving goals.

Why Do We Need AI?

Modern problems have become too complex for humans alone.

Examples include:

- Driving vehicles
- Medical diagnosis
- Language translation
- Fraud detection
- Weather prediction
- Space exploration

AI helps solve these problems faster and more accurately.

Everyday Examples of AI

Students use AI every day without realizing it.

Application

AI Used

Google
Search

Smart ranking

YouTube	Video recommendation
Netflix	Movie recommendation
Amazon	Product recommendation
Google Maps	Route optimization
Siri	Voice Assistant
Alexa	Speech Recognition
ChatGPT	Natural Language Processing
Face Unlock	Face Recognition
Spam Filter	Machine Learning

Characteristics of AI

An intelligent system should have the ability to:

1. Learning

Acquire knowledge from experience.

Example:

- Gmail learns which emails are spam.

2. Reasoning

Draw conclusions from available information.

Example:

- Chess AI chooses the best move.

3. Problem Solving

Find solutions efficiently.

Example:

- GPS finds the shortest path.

4. Perception

Understand information using sensors.

Example:

- Self-driving cars detect pedestrians.

5. Natural Language Understanding

Understand human language.

Example:

- ChatGPT answers questions.

6. Decision Making

Choose the best action.

Example:

- Netflix recommends movies.

Goals of Artificial Intelligence

The major goals are:

- Simulate human intelligence
- Automate repetitive tasks
- Improve decision making
- Solve complex problems
- Assist humans
- Build autonomous systems

Types of Intelligence

Human Intelligence

- Emotional
- Creative
- Flexible
- Learns from few examples

Artificial Intelligence

- Data-driven
- Fast calculations
- Accurate
- Limited to trained tasks

AI vs Traditional Programming

Traditional Programming	Artificial Intelligence
Rules are written manually	Learns rules from data
Same input → Same output	Improves with experience
Cannot learn	Learns continuously
Fixed logic	Adaptive logic

Traditional Programming

Input + Program → Output

Example:

Marks = 95

```
if marks > 35:  
    print("Pass")
```

AI Programming

Data + Learning Algorithm

↓

AI Model

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Predictions

Example:

Thousands of cat images are given to the AI.

Instead of writing rules,

the AI learns:

- Shape
- Eyes
- Fur
- Ears

Now it can recognize a new cat image.

Components of AI

Artificial Intelligence consists of several important areas.

Machine Learning (ML)

Allows computers to learn from data.

Example:

Spam email detection.

Deep Learning

Uses large neural networks.

Example:

Image recognition.

Natural Language Processing (NLP)

Helps computers understand human language.

Examples:

- ChatGPT
- Google Translate

Computer Vision

Allows computers to understand images and videos.

Examples:

- Face recognition
- Self-driving cars

Robotics

Combines AI with physical machines.

Examples:

- Industrial robots
- Delivery robots

Expert Systems

Programs that imitate human experts.

Example:

Medical diagnosis systems.

AI Around Us

Healthcare

- Disease prediction
- Medical imaging
- Drug discovery

Education

- Personalized learning
- AI tutors
- Automatic grading

Banking

- Fraud detection
- Credit scoring
- Chatbots

Agriculture

- Crop monitoring
- Disease detection
- Smart irrigation

Transportation

- Self-driving cars
- Traffic prediction
- Route planning

Entertainment

- Movie recommendations
- Music recommendations
- Video generation

Manufacturing

- Predictive maintenance
- Quality inspection
- Industrial robots

Advantages of AI

- Reduces human effort
- Works 24×7
- High accuracy
- Fast decision making

- Automates repetitive tasks
- Handles large amounts of data
- Improves productivity

Limitations of AI

- Expensive to develop
- Requires large datasets
- Cannot think like humans
- No emotions or creativity
- Ethical concerns
- Job displacement in some sectors
- Bias if trained on biased data

AI Timeline (Brief)

Year	Milestone
1950	Alan Turing proposes the Turing Test
1956	John McCarthy coins the term "Artificial Intelligence"
1997	IBM Deep Blue defeats Garry Kasparov in chess
2011	IBM Watson wins <i>Jeopardy!</i>
2012	Deep Learning breakthrough in image recognition
2016	Google DeepMind AlphaGo defeats Lee Sedol
Today	Generative AI (e.g., ChatGPT, image generators, coding assistants)

AI Misconceptions

Myth: AI is the same as robots.

Fact: AI is software that can also be embedded in robots.

Myth: AI will replace all jobs.

Fact: AI automates certain tasks but also creates new roles and works alongside humans.

Myth: AI thinks like humans.

Fact: AI identifies patterns in data; it does not possess consciousness or emotions.